

Course Title	Interactive Data Visualization		
Course Code	DS612	Credit Structure	3-0-2-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	M.Sc(Data Science)		
Prerequisites	Basic understanding of data and statistics		
Course Objectives/ Brief Course Description	The world is increasingly relying on data collection/analysis for the decision making process. Presently, an enormous amount of data gets generated every day. However, human beings have relatively constant perceptual and cognitive abilities. It is always difficult or almost impossible to notice mistakes within a dataset and that makes data visualization particularly important. Data visualization provides a powerful way to communicate data-driven findings and motivate interesting analyses. Effective visual encoding can supplement cognitive calculations with simple perceptual inferences and improve comprehension and decision making process. Good visual representations help engaging more diverse audiences in the process of analytic thinking. In this course, students will learn the foundations of information visualization and design. Lab assignments using Tableau will sharpen their skills in communicating using data. Goal of the course is to explore the important concepts in human perception, decision-making, color theory, and storytelling as the students get hands-on-experience while applying the concepts in lab assignments / course-project. Short assignments will help students get comfortable with the various techniques we discuss during lectures. The assignments will be done using a real-dataset. Project will give an opportunity to students to apply visualization to a dataset of their choosing.		
Evaluation/ Grading Policy	<i>Exam:</i> One Mid-Sem and end-sem examination. (20 + 35 marks) Lab Assignments: There will be around 6-7 Lab assignments in the whole course comprising a total 30 marks. Lab attendance is compulsory. Lab attendance will carry 20% marks in each assignment. 1 course Project: 15 marks (Project will be evaluated in multiple ways including project presentations of all students). Grading scheme is relative and depends on both: class performance and minimum expectation from a student.		
Course Materials/ References	1. Visualization Analysis and Design by Tamara Munzner 2. Data Visualization by Kieran Healy (https://socviz.co/) 3. Edward R. Tufte , The Visual Display of Quantitative Information 4. Tableau (www.tableau.com/learn/training/20222) 5. B. Shneiderman. The eyes have it: a task by data type taxonomy for information visualizations.		

Detailed Course Content	APPENDIX Instructor: Prof. Bhaskar Chaudhury (bhaskar_chaudhury@daict.ac.in)
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Course Outcome:

CO 1: Understand the fundamental design principles and different types of data visualization

CO 2: Demonstrate the best practice that presents a story in the process of creating data visualization

CO 3: Practice the core principles of data visualization by engaging in case-studies, lab-assignments and course-project using Tableau (widely used interactive data visualization software)

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X		X	X				X	X		X

APPENDIX: Detailed Course Content (Module-wise)

Module 1	Introduction to Data Visualization and Getting Started with Tableau, what and why, perception and cognition,
Module 2	Data Abstraction - Data types, Dataset Types (Tables, fields etc), Attribute Types (Categorical, Ordered etc), Semantics
Module 3	Task abstraction - Actions and Targets, Analyzing and Deriving
Module 4	Marks and Channels - Using Marks and Channels, Visual Channels and encoding, and Channel Effectiveness.
Module 5	Arrangements by Keys and Values, Express Quantitative Values, Separate, Order, Align, Spatial Axis Orientation, Spatial Layout Density.
Module 6	Arrange Spatial fields - Geometry, Geographic Data, Scalar Fields, Vector Fields.
Module 7	Colors - Theory, HSL, colormaps, categorical vs ordered color, color palettes, color deficiency.
Module 8	Manipulate View: Change View over Time, Select Elements, Navigate-facet into Multiple Views. Reduce Items and Attributes, hierarchical and multivariate data.
Module 9	Data visualization choices, design principles, presentation and storytelling techniques using multiple visualization idioms such as Scatterplot, bar and line chart, streamgraph, stacked bar chart, Gantt charts, slope graphs, heatmap, histogram, box-plot, radar plot, pie and coxcomb chart, Sankey diagram, ridgeline plot, radar plot, glyph maps, SPLOM, parallel

	coordinates, choropleth, symbol maps, cartograms, dot density maps, isosurfaces, tooltips, small multiples, MAUP and more.
Case Studies	Anscombe's quartet, John Snow's work on the cholera outbreaks, Charles Minard's chart, visualizing distributions, Bias in Visualization - Misleading visuals, data bias, Uncertainty Visualization, animated bubble charts such as Gapminder Tools and many more for data analysis and storytelling.

Based on availability of time and how the courses progresses, data-visualization based thematic paper discussion are also included in the lecture-sessions.